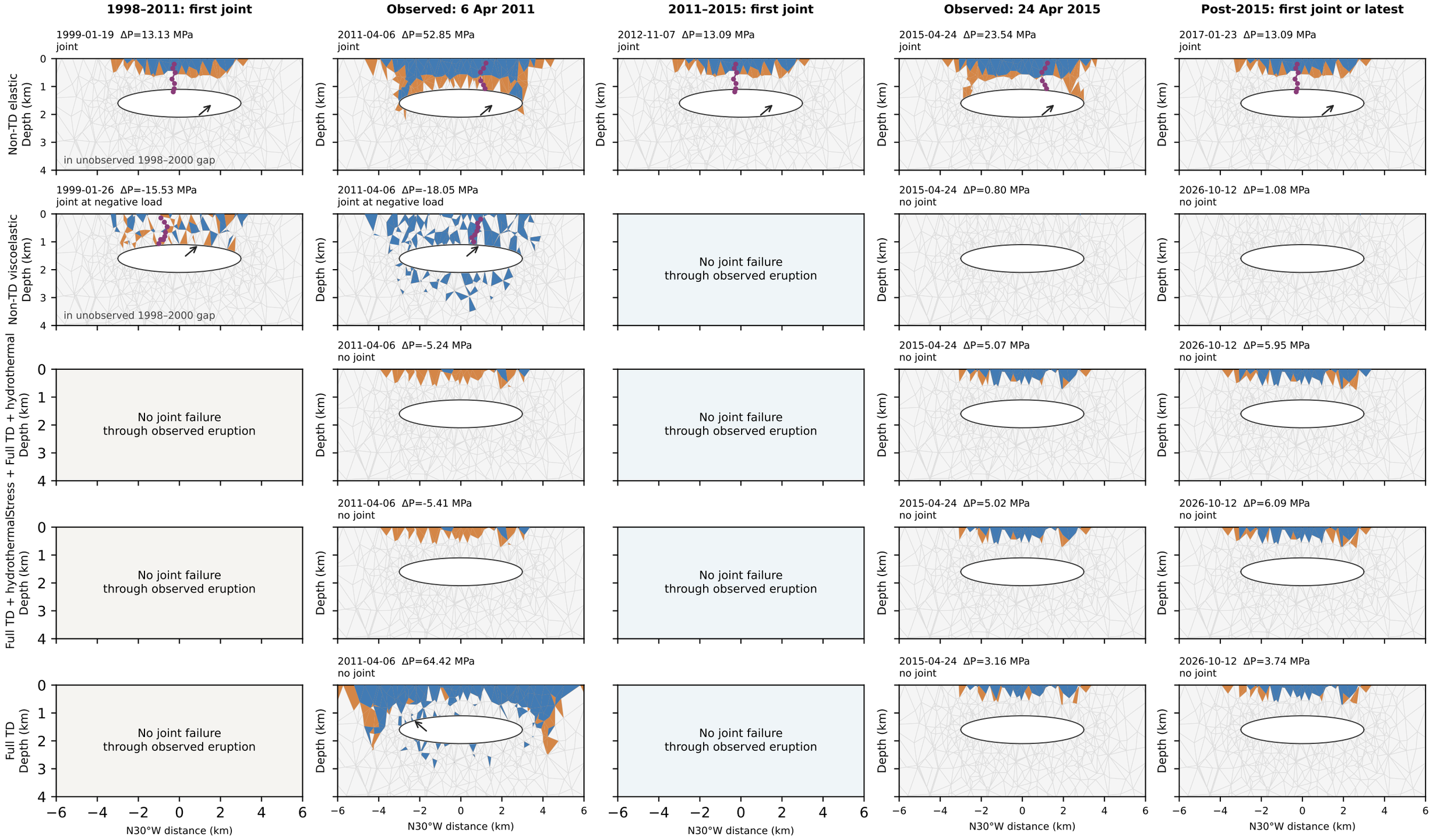

1998–2011: first joint
Observed: 6 Apr 2011
2011–2015: first joint
Observed: 24 Apr 2015
Post-2015: first joint or latest
Non-TD elastic
Non-TD viscoelastic
Full TD + hydrothermal
Full TD + hydrothermalStress + Full TD + hydrothermal
Full TD
Depth (km)
1999-01-19 ΔP=13.13 MPa
joint
2011-04-06 ΔP=52.85 MPa
joint
2012-11-07 ΔP=13.09 MPa
joint
2015-04-24 ΔP=23.54 MPa
joint
2017-01-23 ΔP=13.09 MPa
joint
in unobserved 1998–2000 gap
1999-01-26 ΔP=-15.53 MPa
joint at negative load
2011-04-06 ΔP=-18.05 MPa
joint at negative load
2015-04-24 ΔP=0.80 MPa
no joint
2026-10-12 ΔP=1.08 MPa
no joint
No joint failure
through observed eruption
2011-04-06 ΔP=-5.24 MPa
no joint
2015-04-24 ΔP=5.07 MPa
no joint
2026-10-12 ΔP=5.95 MPa
no joint
2011-04-06 ΔP=-5.41 MPa
no joint
2015-04-24 ΔP=5.02 MPa
no joint
2026-10-12 ΔP=6.09 MPa
no joint
2011-04-06 ΔP=64.42 MPa
no joint
2015-04-24 ΔP=3.16 MPa
no joint
2026-10-12 ΔP=3.74 MPa
no joint
N30°W distance (km)
Mohr–Coulomb yield
Principal tension
3D shear path
First cavity tensile cell (projected)